Introduction to Database System

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Book Reference

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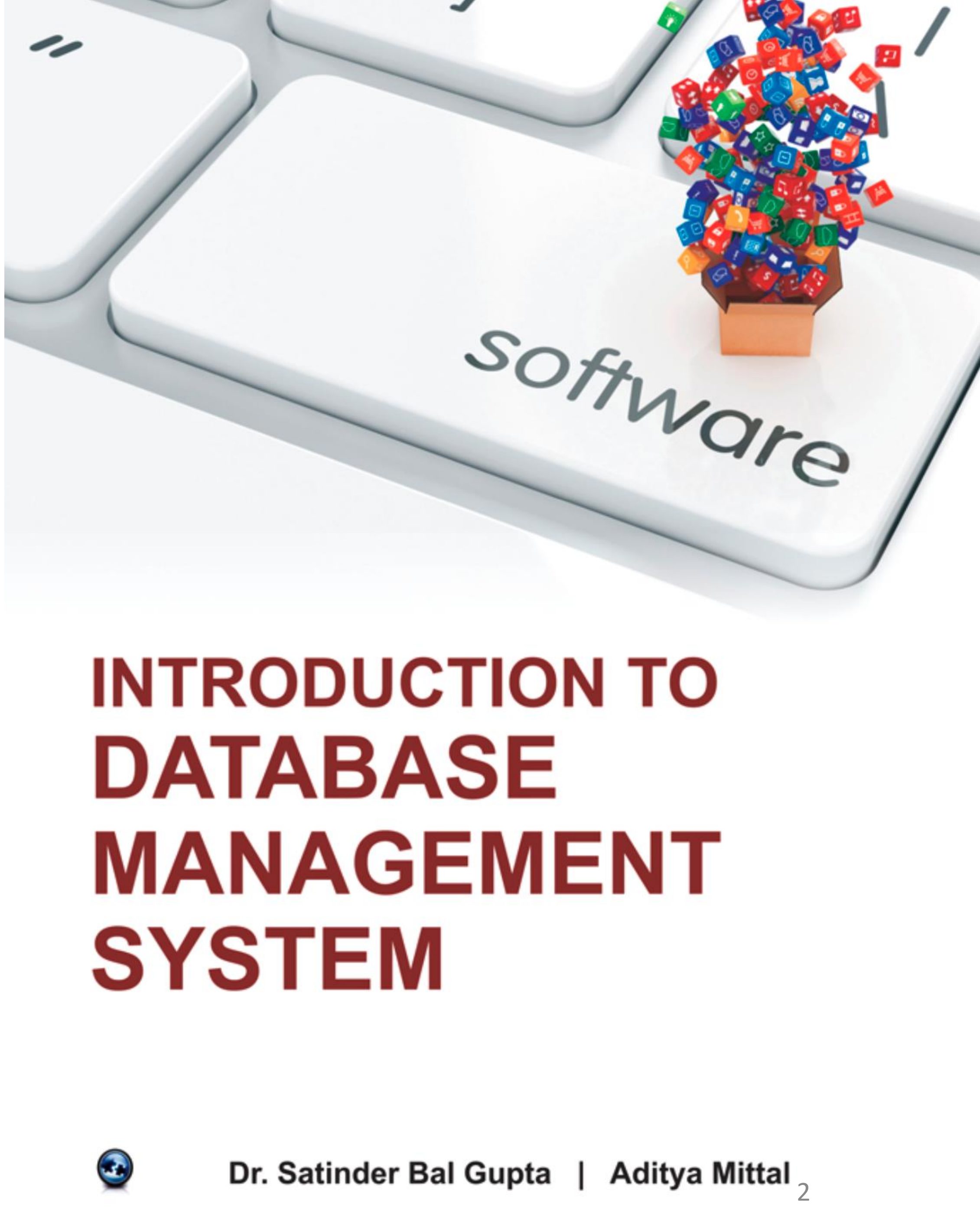

Introduction:

- A Data Base Management System is a software, system or program that allows access to data contained in a database.
- The objective of the DBMS is to provide a convenient and effective method of defining, storing, and retrieving the information stored in the database.

Basic definition and concepts:

- **Data**
  - Defined as known facts that could be recorded and stored on Computer Media
  - It is also defined as raw facts from which the required information is produced
- **Information**
  - Data and information are closely related and are often used interchangeably.
  - Information is nothing but **refined data**.
  - Information consists of data, images, text, documents and voice, but always in a meaningful

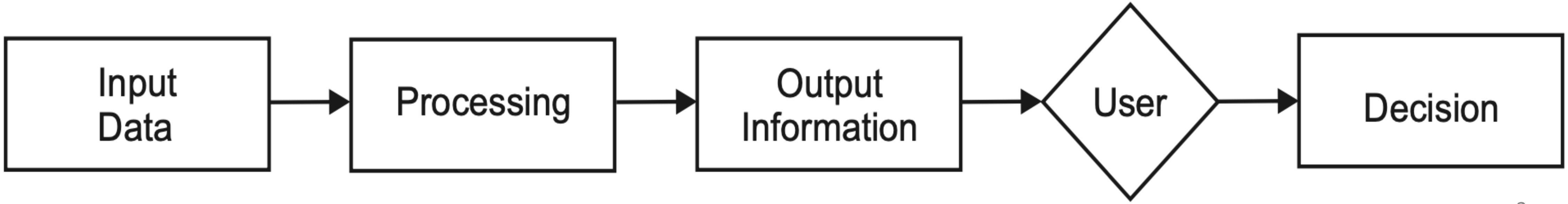

Figure 1: data processing

Information Attribute:

Three major attribute of information

- Accuracy
  - free from errors, and it clearly and accurately reflects the meaning of data
  - free from bias and conveys an accurate picture to the recipient

2. Timeliness
   - recipients receive the information when they need it and within the required time frame.

3. Relevancy
   - usefulness of the piece of information for the corresponding persons.
   - Some information that is relevant for one person might not be relevant for another and vice versa e.g., the price of printer is irrelevant for a rickshaw puller.

Data and Information

Serial Number	**Data**	**Information**
1.	Data is unorganised and unrefined facts	Information comprises processed, organised data presented in a meaningful context
2.	Data is an individual unit that contains raw materials which do not carry any specific meaning.	Information is a group of data that collectively carries a logical meaning.
3.	Data doesn’t depend on information.	Information depends on data.
4.	Raw data alone is insufficient for decision making	Information is sufficient for decision making
5.	An example of data is a student’s test score	The average score of a class is the information derived from the given data.

Meta Data:

- Data about the data.
- Describe objects in the database and makes easier for those objects to be accessed or manipulated.
- Represent the database structure, sizes of data types, constraints, applications, authorization etc., that are used as an integral tool for information resource management.

There are three main types of meta data.

1. Descriptive meta data : It describes a **resource for purpose** such as discovery and identification. In a traditional library cataloging that is form of meta data, title, abstract, author and keywords are examples of meta data.

2. Structural meta data : It describes how compound objects are put together. The example is how pages are ordered to form chapters.

3. Administrative meta data : It provides information to help **manage a resource**, such as when and how it was created, file type and other technical information, and who can access it.

Database:

- ➢ Collection of interrelated data stored together with controlled redundancy to serve one or more applications in an optimal way
- ➢ Collection of logically related data stored together that is designed to meet information requirements of an organization.
- ➢ Databases are organized by fields, records and files.

1. Fields
   - ➢ Smallest unit of the data and called data item or data element.
     - ➢ Ex. Stu_Id, Stu_Name, Address and Telephone number
   - ➢ Represented in the database by a value

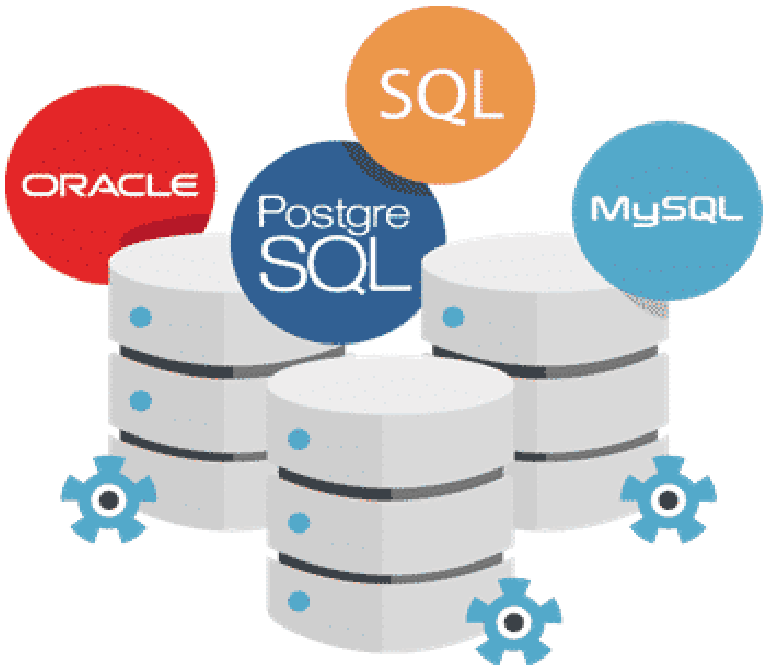

Figure 2: Database

2. Records
   - ➢ collection of logically related fields and each field is possessing a fixed number of bytes and is of fixed data type.
   - ➢ One complete set of fields and each field have some value.
   - ➢ Mainly two types: fixed length records and variable length records.

3. Files
   - ➢ collection of related records.
   - ➢ The records in a file may be of fixed length or variable length depending upon the size of the records contained in a file.

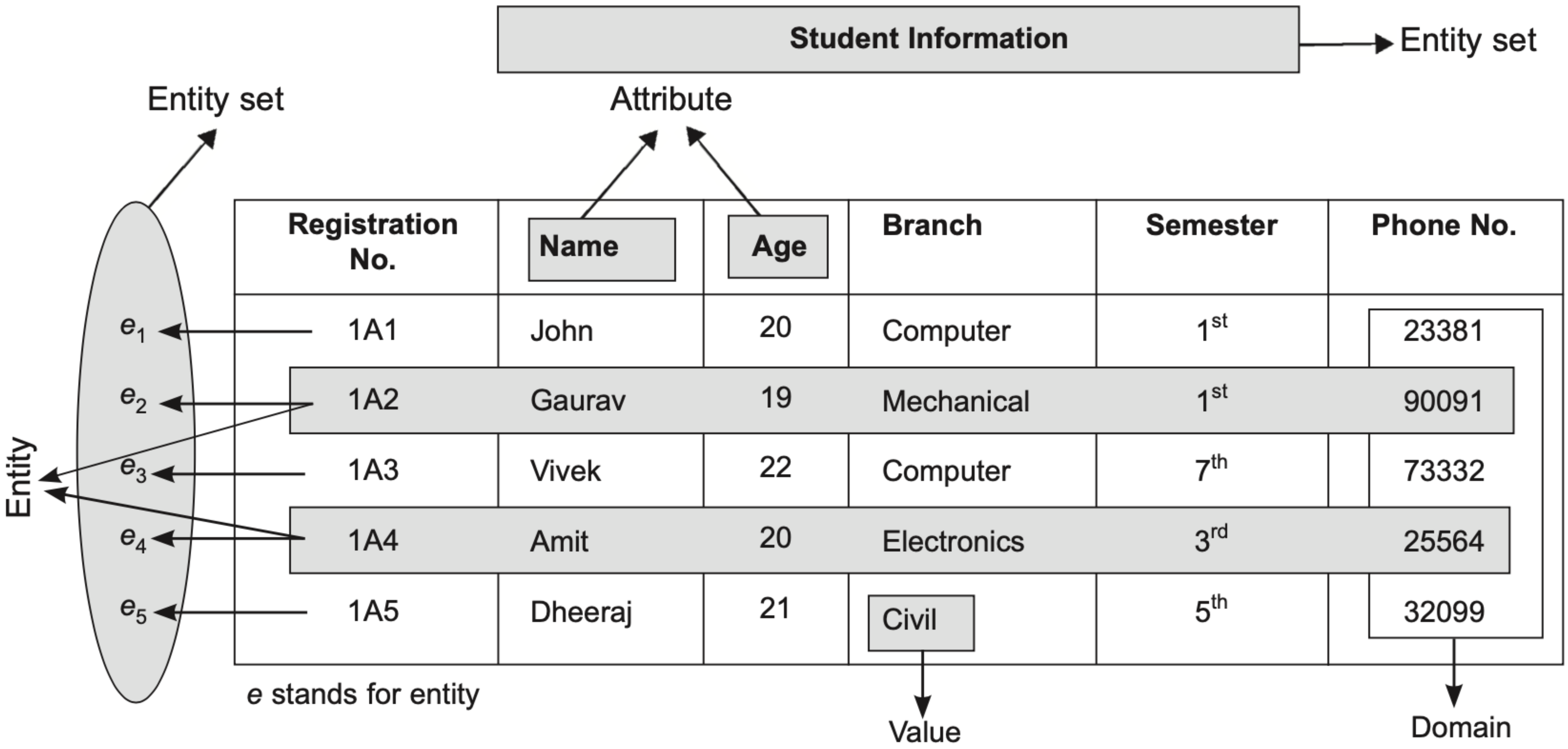

Figure 3: Field, records, and file of a database

Database Management System (DBMS):

- A program or group of programs that work in conjunction with the operating system to **create, process, store, retrieve, control and manage** the data.
- A computerized record-keeping system that stores information and allows the users to **add, delete, modify, retrieve and update** that information.

The DBMS performs the following five primary functions :

1. Define, create and organize a database : The DBMS establishes the logical relationships among different data elements in a database and also defines schemas and subschemas using the DDL.

2. Input data : It performs the function of entering the data into the database through an input device (like data screen, or voice activated system) with the help of the user.

3. Process data : It performs the function of manipulation and processing of the data stored in the database using the DML.

4. Maintain data integrity and security : It allows limited access of the database to authorized users to maintain data integrity and security.

5. Query database : It provides information to the decision makers that they need to make important decisions. This information is provided by querying the database using SQL.

Components of database:

- Three main components of DBMS are Data Definition Language (DDL), Data Manipulation Language and Query Facilities (DML/SQL), and software for controlled access of Database.

1. Data Definition Language (DDL): It allows the users to define the database, specify the data types, data structures and the constraints on the data to be stored in the database.

2. Data Manipulation Language (DML)and Query Language: DML allows users to insert, update, delete and retrieve data from the database.

3. Software for Controlled Access of Database: This software provides the facility of

- controlled access of the database by the users,
- concurrency control to allow shared access of the database and
- recovery control system to restore the database in case of hardware or software failure.

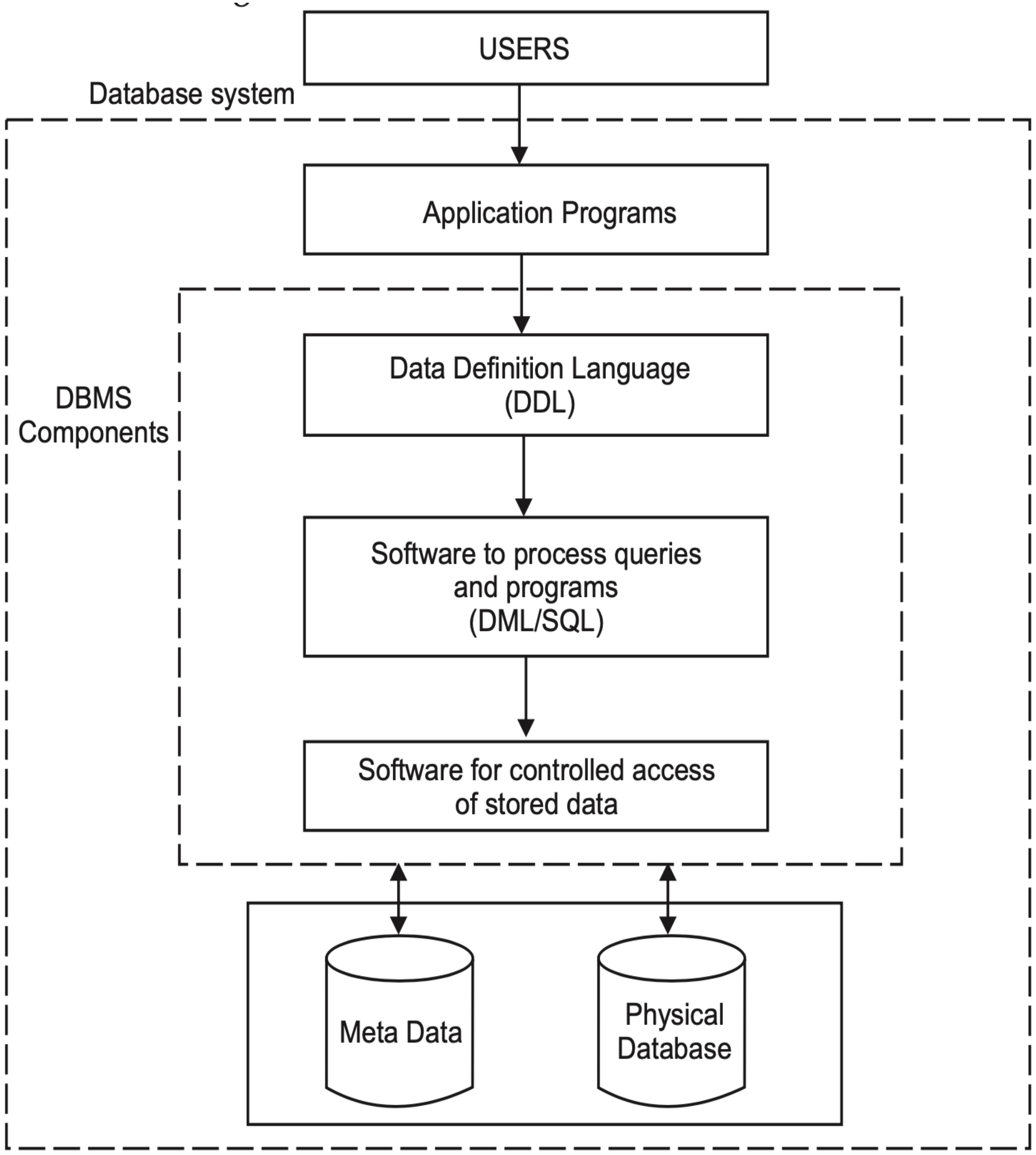

Figure 4: Components of DBMS

Disadvantages of traditional file system

1. Data Redundancy
   - It means the same data may have to be recorded and stored in many files.
   - Ex. personal file and payroll file, both contain data on employee name, designation etc.
   - The result is unnecessary duplicate or redundant data items.

2. Data Inconsistency
   - Data redundancy leads to data inconsistency especially when data is to be updated.
   - Data inconsistency occurs due to the same data items that appear in more than one file do not get updated simultaneously in each and every file.
   - Ex. an employee is promoted from Clerk to Superintendent and the same is immediately updated in the payroll file may not necessarily be updated in provident fund file. This results in two different designations of an employee at the same time leads inconsistency.

3. Lack of Data Integration
   - Due to many independent files, users face difficulty in getting information on any ad hoc query that requires accessing the data stored in many files.
   - A complicated programs have to be developed to retrieve data from every file or the users have to manually collect the required information.

Disadvantages of traditional file system

4. Program Dependence

- The reports produced by the file processing system are program dependent
- If any change in the format or structure of data and records in the file is to be made, the programs have to modified correspondingly.

5. Data Dependence

- The Applications/programs in file processing system are data dependent
- The file organization, its physical location and retrieval from the storage media are dictated by the requirements of the particular application.
- For example, in payroll application, the file may be organized on employee records sorted on their last name, which implies that accessing of any employee's record has to be through the last name only.

6. Limited Data Sharing

- There is limited data sharing possibilities with the traditional file system.
- Each application has its own private files and users have little choice to share the data outside their own applications.
- Complex programs required to be written to obtain data from several incompatible files.

Disadvantages of traditional file system

7. Poor Data Control

- There was no centralized control at the data element level, hence a traditional file system is decentralized in nature.

8. Problem of Security

- It is very difficult to enforce security checks and access rights in a traditional file system, since application programs are added in an ad-hoc manner.

9. Data Manipulation Capability is Inadequate

- The data manipulation capability is very limited in traditional file systems since they do not provide strong relationships between data in different files.

10. Needs Excessive Programming

- An excessive programming effort was needed to develop a new application program due to very high interdependence between program and data in a file system.

Advantages of database management systems:

1. Controlled redundancy : In a database system, this duplication is carefully controlled, that means the database system is aware of the redundancy and it assumes the responsibility for propagating updates.

2. Data consistency : The problem of updating multiple files in traditional file system leads to inaccurate data as different files may contain different information of the same data item at a given point of time. This causes incorrect or contradictory information to its users. In database systems, this problem of inconsistent data is automatically solved by controlling the redundancy.

3. Program data independence : The traditional file systems are generally data dependent, which implies that the data organization and access strategies are dictated by the needs of the specific application and the application programs are developed accordingly. However, the database systems provide an independence between the file system and application program, that allows for changes at one level of the data without affecting others.

4. Sharing of data : In database systems, the data is centrally controlled and can be shared by all authorized users. The sharing of data means not only the existing applications programs can also share the data in the database but new application programs can be developed to operate on the existing data.

5. Enforcement of standards : In database systems, data being stored at one central place, standards can easily be enforced by the DBA. This ensures standardized data formats to facilitate data transfers between systems.

6. Improved data integrity : Data integrity means that the data contained in the database is both accurate and consistent. The centralized control property allow adequate checks can be incorporated to provide data integrity.

7. Improved security : Database security means protecting the data contained in the database from unauthorized users. The DBA ensures that proper access procedures are followed, including proper authentical schemes for access to the DBMS and additional checks before permitting access to sensitive data.

8. Data access is efficient : The database system utilizes different sophisticated techniques to access the stored data very efficiently.

9. Conflicting requirements can be balanced : The DBA resolves the conflicting requirements of various users and applications by knowing the overall requirements of the organization.

10. Improved backup and recovery facility : Through its backup and recovery subsystem, the database system provides the facilities for recovering from hardware or software failures.

11. Minimal program maintenance : In a traditional file system, the application programs with the description of data and the logic for accessing the data are built individually. These are reduced to minimal in database systems due to independence of data and application programs.

12. Data quality is high : The quality of data in database systems are very high as compared to traditional file systems. This is possible due to the presence of tools and processes in the database system.

13. Good data accessibility and responsiveness : The database systems provide query languages or report writers that allow the users to ask ad hoc queries to obtain the needed information immediately, without the requirement to write application programs (as in case of file system), that access the information from the database.

14. Concurrency control : The database systems are designed to manage simultaneous (concurrent) access of the database by many users.

15. Economical to scale : In database systems, the operational data of an organization is stored in a central database. The application programs that work on this data can be built with very less cost as compared to traditional file system.

16. Increased programmer productivity : The database system provides many standard functions that the programmer would generally have to write in file system. The availability of these functions allow the programmers to concentrate on the specific functionality required by the users without worrying about the implementation details. This increases the overall productivity of the programmer and also reduces the development time and cost.

Disadvantages of DBMS:

1. Complexity increases : The data structure may become more complex because of the centralized database supporting many applications in an organization.

2. Requirement of more disk space : The wide functionality and more complexity increase the size of DBMS. Thus, it requires much more space to store and run than the traditional file system.

3. Additional cost of hardware : The cost of database system's installation is much more. It depends on environment and functionality, size of the hardware and maintenance costs of hardware.

4. Cost of conversion : The cost of conversion from old file-system to new database system is very high.

5. Need of additional and specialized manpower : Any organization having database systems, need to be hire and train its manpower on regular basis to design and implement databases and to provide database administration services.

6. Need for backup and recovery : For a database system to be accurate and available all times, a procedure is required to be developed and used for providing backup copies to all its users when damage occurs.

7. Organizational conflict : A centralized and shared database system requires a consensus on data definitions and ownership as well as responsibilities for accurate data maintenance.

8. More installation and management cost : The big and complete database systems are more costly. They require trained manpower to operate the system and has additional annual maintenance and support costs.

Database Systems or Database System environment:

The DBMS software together with the Database is called a database system. In other words, it can be defined as an organization of components that define and regulate the collection, storage, management and use of data in a database.

1. Data
   - The whole data in the system is stored in a single database.
   - This data in the database are both shared and integrated.
   - Sharing of data means individual pieces of data in the database is shared among different users

2. Hardware
   - Hardware consists of the secondary storage devices like disks, drums where the database stored.
   - There is two types of hardware. The first one is processor and main memory that supports in running the DBMS.
   - The second one is the secondary storage devices, i.e., hard disk, magnetic disk etc., that are used to hold the stored data.

3. Software
   - A layer or interface of software exists between the physical database and the users. This layer is called the DBMS.

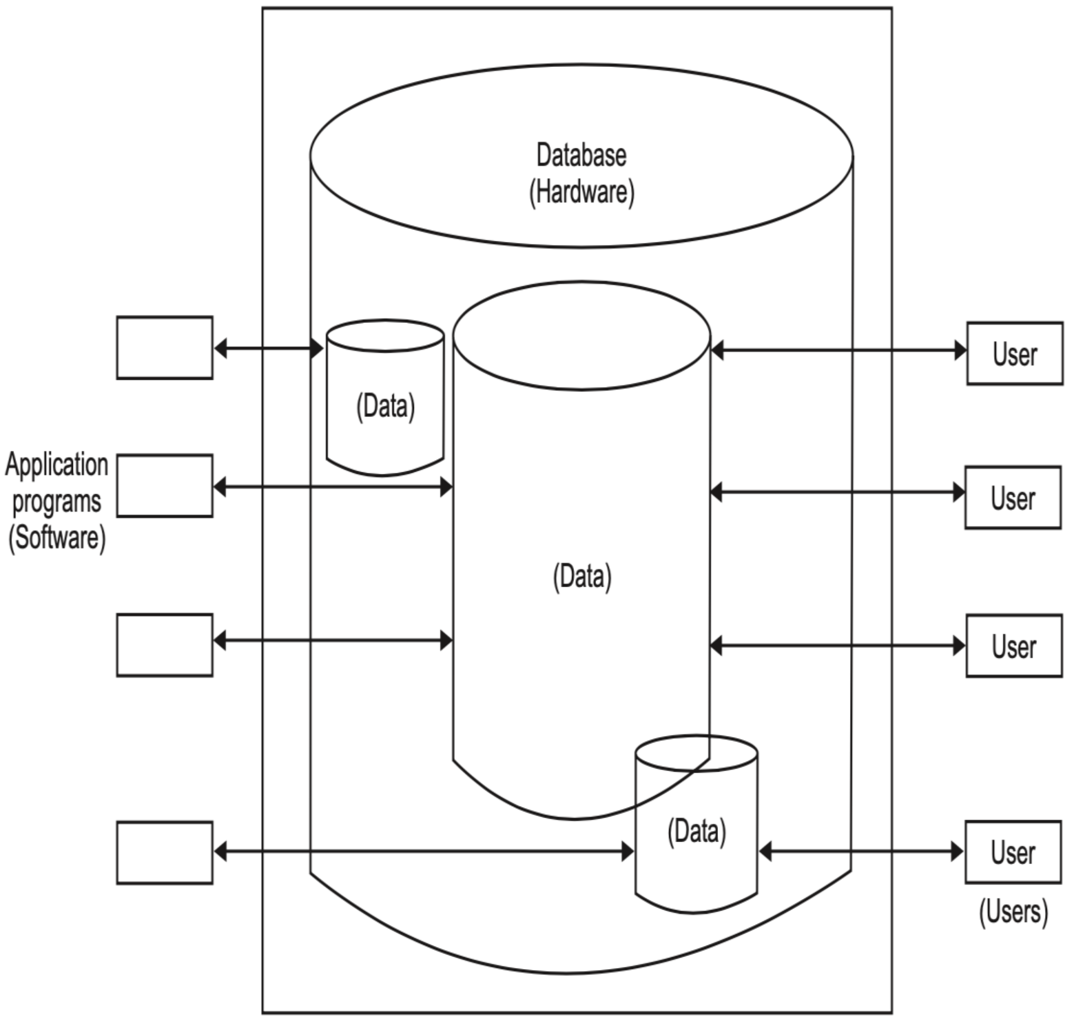

Figure 5: Database system

4. Users

- The users are the people interacting with the database system.
- There are four types of users interacting with the database systems. These are Application Programmers, online users, end users or naive users and finally the Database Administrator (DBA).

DBMS Users:

1. End users or Naive users: The end users or naive users use the database system through a menu-oriented application program, where the type and range of response is always displayed on the screen. A user of an ATM falls in this category.

2. Online users: These type of users communicate with the database directly through an online terminal or indirectly through an application program and user interface.

3. Application Programmers: These are the professional programmers or software developers who develop the application programs or user interfaces for the naive and online users. These programmers must have the knowledge of programming languages such as Assembly, C, C++, Java, or SQL, etc., since the application programs are written in these languages.

4. Database Administrator: DBA is a person who have complete control over database of any enterprise.. He is concerned with the Back-End of any project.

Database administrator:

Database Administrator (DBA) is a person who have complete control over database of any enterprise.. He is concerned with the Back-End of any project. Some of the main responsibilities of DBA are as follows :

1. Deciding the conceptual schema or contents of database : DBA decides the data fields, tables, queries, data types, attributes, relations, entities or you can say that he is responsible for overall logical design of database.

2. Deciding the internal schema of structure of physical storage : DBA decides how the data is actually stored at physical storage, how data is represented at physical storage(physical environment).

3. Deciding users : DBA gives permission to users to use database. Without having proper permission, no one can access data from database.

4. Deciding user view : DBA decides different views for different users.

5. Granting of authorities : DBA decides which user can use which portion of database. User can use only that data on which access right is granted to him.

6. Deciding constraints : DBA decides various constraints over database for maintaining consistency and validity in database.

7. Security : Security is the major concern in database. DBA takes various steps to make data more secure against various disasters and unauthorized access of data.

8. Monitoring the performance : DBA is responsible for overall performance of database. DBA regularly monitors the database to maintain its performance and try to improve it.

9. Backup : DBA takes regular backup of database, so that it can be used during system failure. Backup is also used for checking data for consistency.

10. Removal of dump and maintain free space : DBA is responsible for removing unnecessary data from storage and maintain enough free space for daily operations. He can also increase storage capacity when necessary.

11. Checks : DBA also decides various security and validation checks over database to ensure consistency.

12. Liaisioning with users : Another task of the DBA is to liaisioning with users and ensure the availability of the data they require and write the necessary external schemas.

DBMS languages:

The following five languages are available to specify different schemas.

1. Data Definition Language (DDL)
2. Storage Definition Language (SDL)
3. View Definition Language (VDL)
4. Data Manipulation Language (DML)
5. Fourth-Generation Language (4-GL)

1. Data Definition Language (DDL): It is used to specify a database conceptual schema using set of definitions.

2. Storage Definition Language (SDL): It is used to specify the internal schema in the database. The storage structure and access methods used by the database system is specified by the specified set of SDL statements.

3. View Definition Language (VDL): It is used to specify user's views and their mappings to the conceptual schema. But generally, DDL is used to specify both conceptual and external schemas in many DBMS's.

4. Data Manipulation Language (DML): It provides a set of operations to support the basic data manipulation operations on the data held in the database. It is used to query, update or retrieve data stored in a database.

The DML is of the two types :

(i) Procedural DML : It allows the user to tell the system what data is needed and how to retrieve it.
(ii) Non-procedural DML : It allows the user to state what data are needed, rather than how it is to be retrieved.

Fourth generation language (4GL):

It is a compact, efficient and non-procedural programming language used to improve the efficiency and productivity of the DBMS. The 4-GL has the following components in it. These are :

(a) Query languages
(b) Report
(c) Spread sheets
(d) Database languages
(e) Application generators
(f) High level languages to generate application program.

Schemas, Subschema and Instances:

- Schema:
  - A schema is plan of the database that give the names of the entities and attributes and the relationship among them.
  - It is defined as a framework into which the values of the data items are fitted. The values fitted into the framework changes regularly but the format of schema remains the same e.g., consider the database consisting of three files ITEM, CUSTOMER and SALES. The data structure diagram for this schema is shown below. The schema is shown in database language.

```
Schema name is  ITEM_SALES_REC

        type    ITEM = record
                ITEM_ID: string;
                ITEM_DESC: string;          } Attributes/data items
                ITEM_COST: integer;
                end
        type    CUSTOMER = record
                CUSTOMER_ID = integer;
                CUSTOMER_NAME = string;
                CUSTOMER_ADD = string;
                CUSTOMER_CITY = string;
                CUSTOMER_BAL = integer;
                end
        type    SALES = RECORD
                CUSTOMER_ID = integer;
                ITEM_ID = string;
                ITEM_QTY = integer;
                ITEM_PRICE = integer;
                end
```

Figure 6: Data structure diagram for the item sales record.

Generally, a schema can be partitioned into two categories,

(i) Logical schema

- The logical schema is concerned with exploiting the data structures offered by the DBMS so that the schema becomes understandable to the computer.

(ii) physical schema

- ➢ The physical schema is concerned with the manner in which the conceptual database get represented in the computer as a stored database. It is hidden behind the logical schema and can usually be modified without affecting the application programs.

- Subscema
  - ➢ A subschema is a subset of the schema having the same properties that a schema has.
  - ➢ The subschema allows the user to view only that part of the database that is of interest to him.

- Instances

➢ The data in the database at a particular moment of time is called an instance or a database state.

➢ In a given instance, each schema construct has its own current set of instances. The Figure below shows an instance of the ITEM relation in a database schema.

ITEM

ITEM-ID	ITEM_DESC	ITEM_COST
1111A	Nutt	3
1112A	Bolt	5
1113A	Belt	100
1144B	Screw	2

Figure 7: An instance/database state of the ITEM relation.

Three level architecture of database systems (dbms):

The architecture is a framework for describing database concepts and specifying the structure of database system. A database can be divided into three levels: external level, conceptual level and internal level

(i) Internal Level

- ➢ Internal level describes the actual physical storage of data or the way in which the data is actually stored in memory.
- ➢ This level is not relational because data is stored according to various coding schemes instead of tabular form (in tables).

The internal level is concerned with the following aspects:

– Storage space allocation
– Access paths
– Data compression and encryption techniques
– Record placement etc.

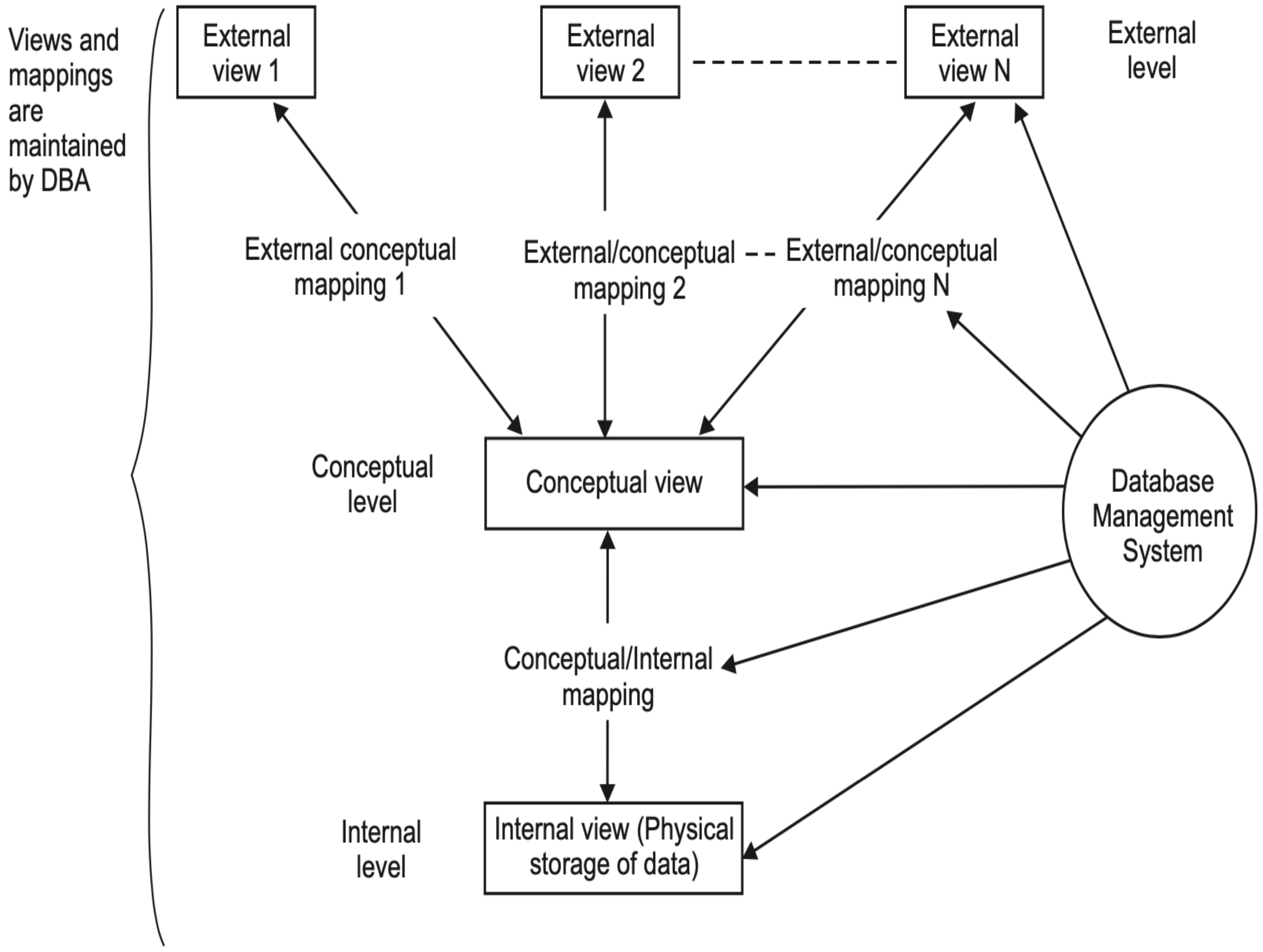

Figure 8: Three level architecture of DBMS.

(ii) Conceptual Level

- The conceptual level is also known as logical level which describes the overall logical structure of whole database for a community of users.
- This level is relational because data visible at this level will be relational tables and operators.
- This level represents entire contents of the database in an abstract form in comparison with physical level

(iii) External Level

- The external level is concerned with individual users.
- This level describes the actual view of data seen by individual users.
- The external schema is defined by the DBA for every user.
- The remaining part of database is hidden from that user.

Data Models:

A data model is a collection of concepts that can be used to describe the structure of the database including data types, relationships and the constraints that apply on the data.

Characteristics of data models:

A data model must posses the following characteristics so that the best possible data representation can be obtained.
(i) Diagrammatic representation of the data model.
(ii) Simplicity in designing i.e., Data and their relationships can be expressed and distinguished easily.
(iii) Application independent, so that different applications can share it.
(iv) Data representation must be without duplication.
(v) Bottom-up approach must be followed.
(vi) Consistency and structure validation must be maintained.

Types of data models:

The various data models can be divided into three categories, such as,

(i) Record Based Data Models : These models represent data by using the **record structures**. These data models can be further categorised into three types:

(a) Hierarchical Data Model
(b) Network Data Model
(c) Relational Data Model.

(ii) Object Based Data Models : These models are used in describing the data at the **logical and user view** levels. These data models can be further categorised into four types:

(a) Entity Relationship Model (ER-Model)
(b) Object Oriented Model
(c) Semantic Data Model
(d) Functional Data Model. The models are discussed in the coming sections.

(iii) Physical Data Models : These models provide the concepts that describes the details of how the data is stored in the computer along with their record structures, access paths and ordering. These data models can be divided into two types:

(a) Unifying Model.

(b) Frame Memory Model.

Record Based Data Models:

Record based data models represent data by using the record structures. These models lie between the object based data models and the physical data models. The various categories of record based data models are as follows:

- Hierarchical Data Model :
  - The hierarchical data model organizes records in a tree structure i.e., hierarchy of parent and child records relationships.
  - A record is a collection of field values that provide information of an entity.

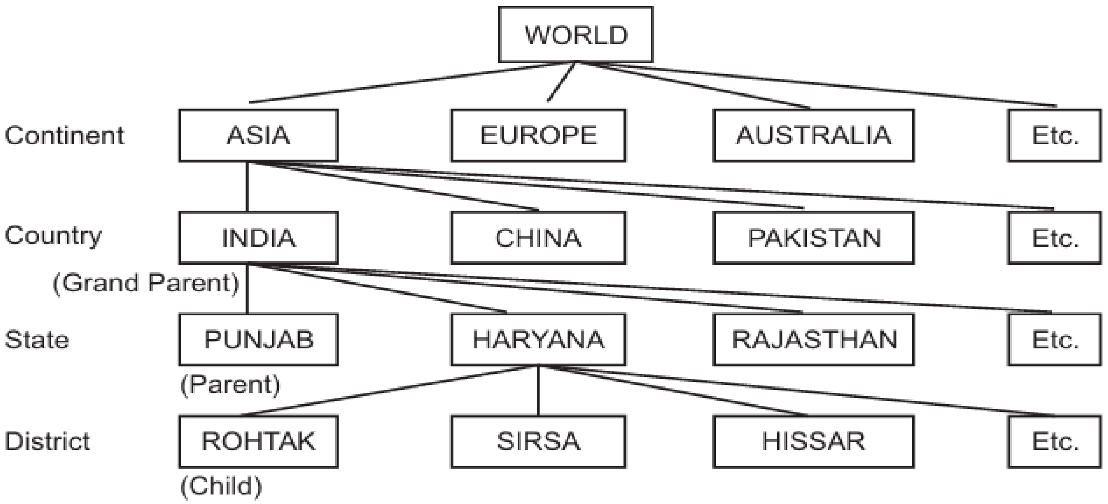

FIGURE 1.8. Hierarchical Model.

- Network Data Model :
  - ➢ Able to handle many to many (N : N) relations between its records is the main distinguishing feature from the hierarchical model.
  - ➢ Thus, this model permits a child record to have more than one parent.
  - ➢ Directed graphs are used instead of tree structure in which a node can have more than one parent.
  - ➢ The relationships between specific records of 1 : 1 (one to one), 1 : N (one to many) or N : N (many to many) are explicitly defined in database definition of this model.
  - ➢ There are two basic data structures in this model—Records and Sets.

- Relational data Model :
  - ➢ Most popular developments in the database technology because it can be used for representing most of the real world objects and the relationships between them.
  - ➢ The main significance of the model is the absolute separation of the logical view and the physical view of the data.
  - ➢ The physical view in relational model is implementation dependent and not further defined.
  - ➢ The logical view of data in relational model is set oriented.
  - ➢ A relational set is an unordered group of items.
  - ➢ The field in the items are the columns. The column in a table have names. The rows are unordered and unnamed.
  - ➢ A database consists of one or more tables plus a catalogue describing the database. The relational model consists of three components:

Object Based Data Models:

- ➢ Known as conceptual models used for defining concepts including entities, attributes and relationships between them.
- ➢ An entity is a distinct object which has existence in real world.
- ➢ It will be implemented as a table in a database.
- ➢ An attribute is the property of an entity, in other words, attribute is a single atomic unit of information that describes something about its entity.

There are 4 types of object based data models. These are:

- **Entity-Relationship (E-R) Model :**
  - Generalization of model like the hierarchical and network model.
  - It also allows the representation of the various constraints as well as their relationships.
  - E-R relationship is of 1 : 1, 1 : N or N : N type which tells the mapping from one entity set to another.
  - E-R model is shown diagrammatically using entity-relationship (E-R) diagrams which represents the elements of the conceptual model.
  - The various features of E-R model are:
    - ✓ E-R Model can be easily converted into relations (tables).
    - ✓ E-R Model is used for purpose of good database design by database developer.
    - ✓ It is helpful as a problem decomposition tool as it shows entities and the relationship between those entities.
    - ✓ It is very simple and easy to understand by various types of users.

- Object-Oriented Data Model :
  - A logical data model that captures the semantics of objects supported in an object-oriented programming.
  - It is based on collection of objects, attributes and relationships which together form the static properties.
  - An object is a collection of data and methods. When different objects of same type are grouped together they form a class.
  - The object model is represented graphically with object diagrams containing object classes.
  - Classes are arranged into hierarchies sharing common structure and behavior and are associated with other classes.

Semantic Data Models :

  - These models are used to express greater interdependencies among entities of interest.
  - This class of data models are influenced by the work done by artificial intelligence researchers.
  - Semantic data models are developed to organize and represent knowledge but not data.
  - Mainframe database are increasingly adopting semantic data models. Also, its growth usage is seen in PC's. In coming times database management systems will be partially or fully intelligent.

- Functional Data Model :
  - The functional data model describes those aspects of a system concerned with transformation of values-functions, mappings, constraints and functional dependencies.

- It shows how output value in computation are derived from input values without regard for the order in which the values are computed. It also includes constraints among values. It consists of multiple data flow diagrams. Traditional computing concepts such as expression trees are examples of functional models.

Which Data Models to use?

Data models are essential as they provide access techniques and data structure for defining a DBMS. In other words, a data model describe the logical Organization of data along with operations that manipulate the data. We have large number of data models, the one which is best for the Organization depends upon the following factors:

- Is the database **too small or too big.**
- What are the **costs** involved.
- The volume of **daily transactions** that will be done.
- The estimated **number of queries** that will be made from the database by the organization to enquire about the data.
- The data requirements of the organization using it.

From the available record based data models, **the relational data model** is most commonly used model by most of the organizations because of the following reasons:

1. It increases the productivity of application programmers in designing the database. Whenever changes are made to the database there is no need of changing the application programs because of separation of logical level from conceptual level.

2. It is useful for representing most of the real world objects and relationships between them.

3. It provides very powerful search, selection and maintenance of data.

4. It hides the physical level details from the end users so end users are not bothered by physical storage.

5. It provides data integrity and security so that data is not accessed by unauthorized users and data is always accurate.

6. It provides adhoc query capability.

Comparison of Various Data Models

The most commonly used data models are compared on the basis of various properties. The comparison table is given below.

Property	Hierarchical	Network	Relational	E-R Diagram	Object-oriented
1. Data element organization	Files, records	Files, records	Tables/tuples	Objects, entity sets	Objects
2. Identity	Record based	Record based	Value based	Value based	Record based
3. Data Independence	Yes	Yes	Yes	Yes	Yes
4. Relationship Organization	Logical proximity in a linearised tree.	Intersecting Networks	Identifiers of rows in one table are embedded as attribute values in another table.	Relational extenders that support specialized applications.	Logical containment
5. Access Language	Procedural	Procedural	Non-procedural	Non-procedural	Procedural
6. Structural Independence	No	No	Yes	Yes	Yes

Types of database systems:

The database systems can be classified into three categories
(i) According to the number of users
(ii) According to the type of use
(iii) According to database site locations

1. According to the Number of users:

According to the number of users, the database systems can be further subdivided into two categories, namely:
(a) Single-user database systems
(b) Multiuser database systems.

Microcomputer or Workstation
User Interface
DBMS
Operating System
Database Files (Storage)
Communications

Figure 9: Singe user database

(a) Single-user database systems :

- he database reside on a PC–on the hard disk.
- All the applications run on the same PC and directly access the database.
- In single user database systems, the application is the DBMS.
- A single user database system is shown in figure 9.

(b) Multiuser database systems :

- many PC's are connected through a Local Area Network (LAN) and a file server stores a copy of the database files.
- Each PC on the LAN is given a volume name on the file server.
- Applications run on each PC that is connected to the LAN and access the same set of files on the file server.
- The applications must handle the concurrency control and the business rules are enforced in the application.
- The example is MS-Access or Oracle files on a file server. A multiuser database system is shown in figure 10.

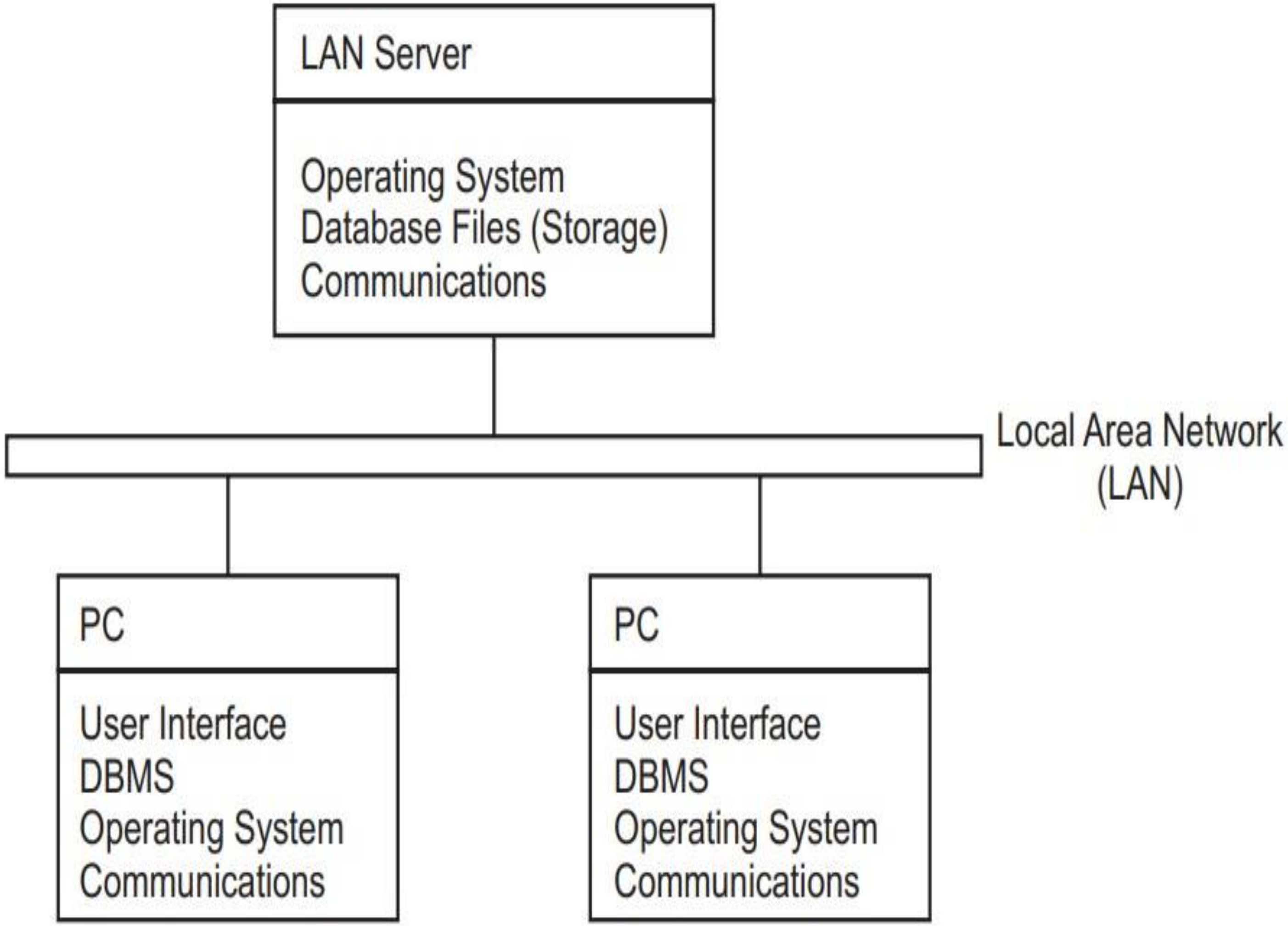

Figure 10: Multi-user database

Advantages of Multiuser Database System

There are many advantages of multiuser database system. Some of them are as follows:
(i) Ability to share data among various users.
(ii) Cost of storage is now divided among various users.
(iii) Low cost since most components are now commodity items.

Disadvantages of Multiuser Database System :

The major disadvantage of the multiuser database system is that it has **a limited data sharing ability** , only a few users can share the data at most.

2. According to the Type of use:

According to the type of use, the database systems can be further subdivided into three categories, namely:
(a) Production or Transactional Database Systems
(b) Decision Support Database Systems
(c) Data Warehouses.

(a) Production or Transactional Database Systems :

- The production database systems are used for management of supply chain and for tracking production of items in factories, inventories of items in warehouses/stores and orders for items.
- The transactional database systems are used for purchases on credit cards and generation of monthly statements.
- They are also used in Banks for customer information, accounts, loans and banking transactions.

(b) Decision Support Database Systems :

- Decision support database systems are interactive, computer-based systems that aid users in judgement and choice activities.
- They provide data storage and retrieval but enhance the traditional information access and retrieval functions with support for model building and model based reasoning.
- They support framing, modelling and problem solving.
- Typical application areas of DSS's are management and planning in business, health care, military and any area in which management will encounter complex decision situations.

(c) Data Warehouses :

- A data warehouse is a relational database management system (RDMS) designed specifically to meet the transaction processing systems.
- It can be loosely defined as any centralized data repository which can be queried for business benefit.

3. According to Database Site Locations:

(a) Centralized database systems
(b) Parallel database systems
(c) Distributed database systems
(d) Client/Server database systems

(a) Centralized database systems :

- It consists of a single processor together with its associated data storage devices and other peripherals.
- Multiple users access the applications through simple terminals that have no processing power of their own.
- The user interface is text-mode screens and the business rules are enforced in the applications running on the mainframe or PC.
- The example of centralized database system is DB2 database and Cobol application programs running on IBM 390.

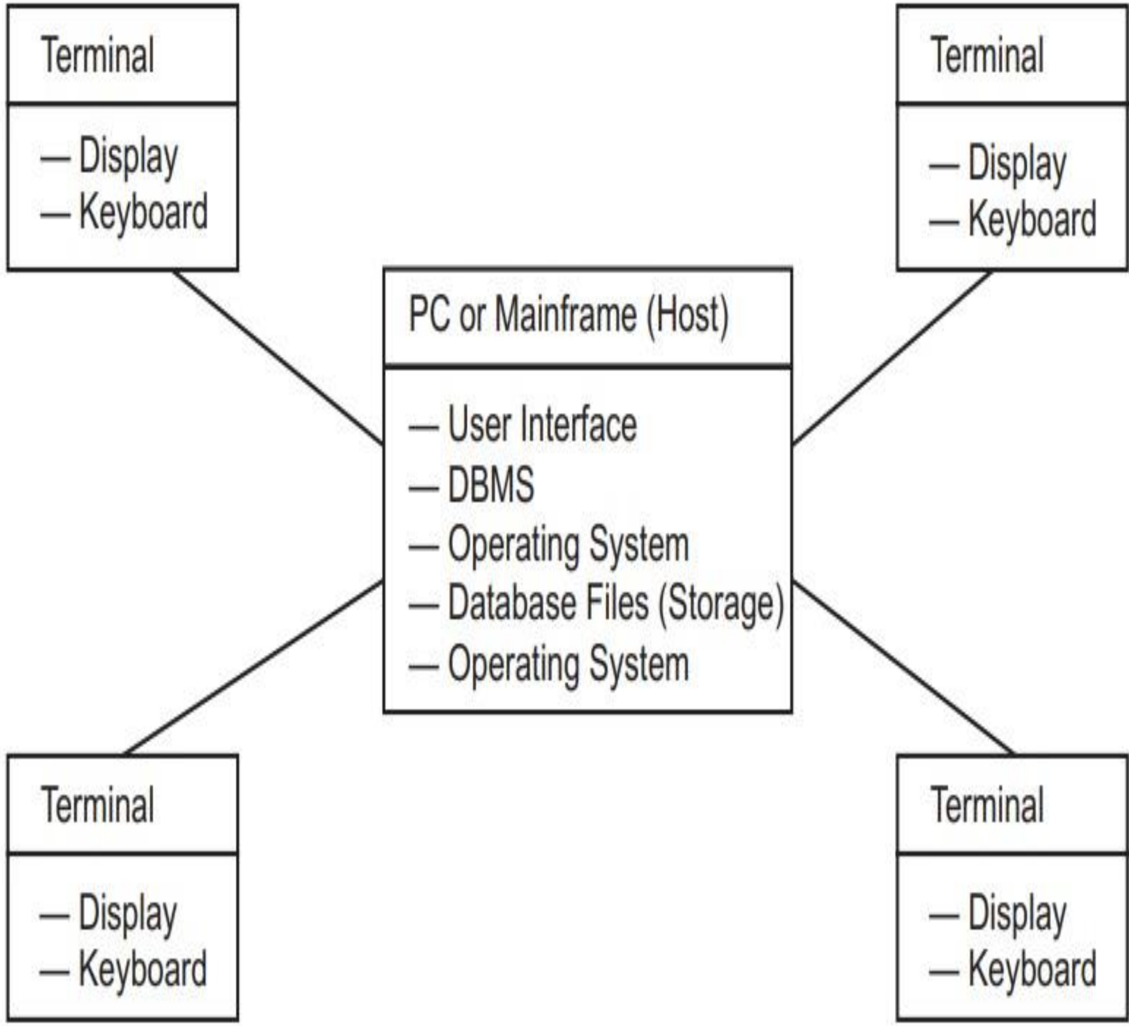

Figure 11: centralized Database system

(b) Parallel database systems :

- A parallel database system can be defined as a database system implemented on a tightly coupled multiprocessor or on a loosely coupled multiprocessor.
- Parallel database systems link multiple smaller machines to achieve the same throughput as a single larger machine, often with greater scalability and reliability than single processor database system.

There are three main architectures for parallel database system.

(i) Shared memory architecture
(ii) Shared disk architecture
(iii) Shared nothing architecture

(c) Distributed database systems :

- Data is spread across a variety of different databases.
- These machines are widely spread and are connected through the communication networks.
- Each machine can have is own data and applications, and can access data stored on other machines.
- Distributed database system is a combination of logically interrelated databases distributed over a computer network and the distributed database management system (DDBMS).

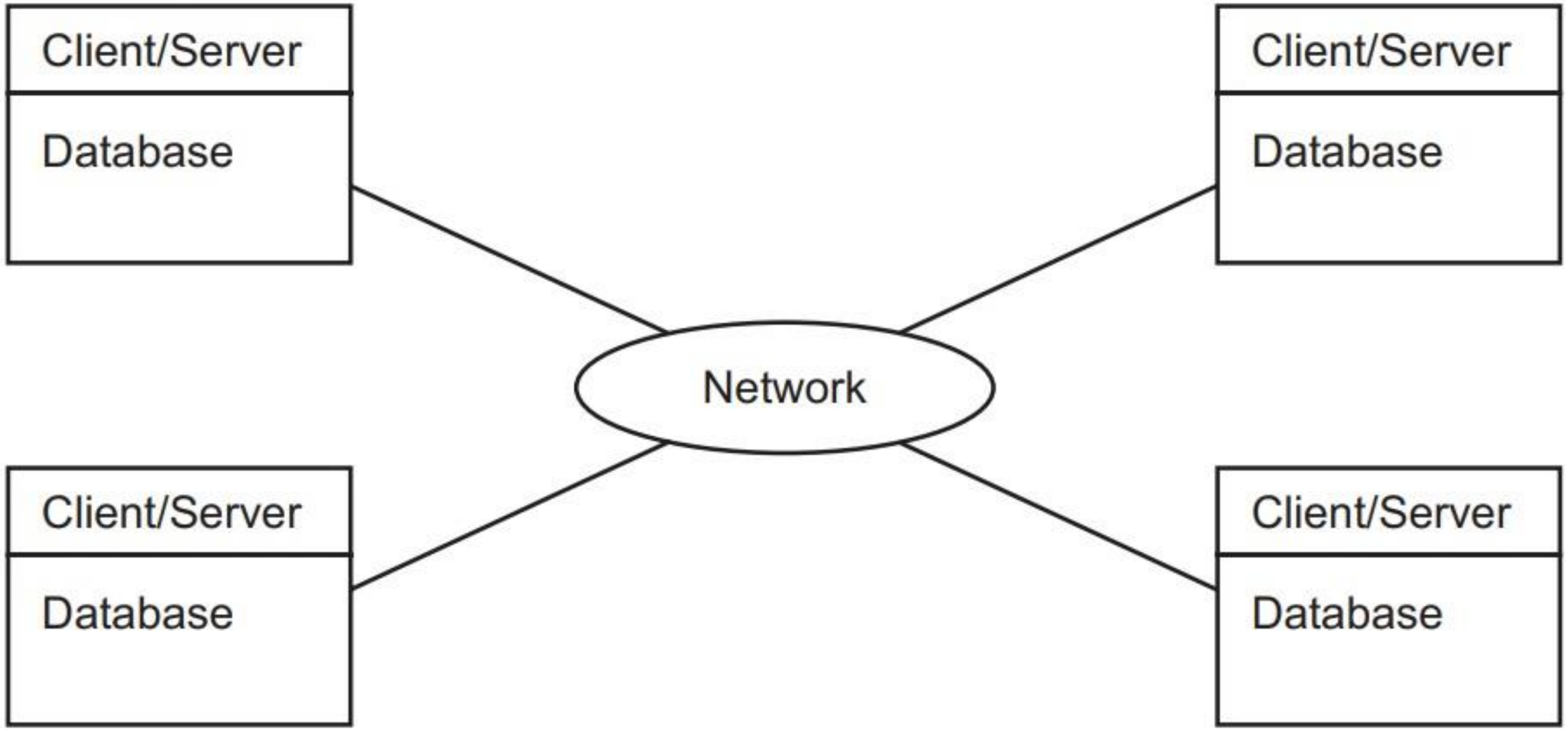

Figure 12: Distributed Database

(d) Client/Server Database System :

- The server acts as a whole data base management system and some clients or personal computers which are connected with server through a network interface. The complete architecture is shown below:

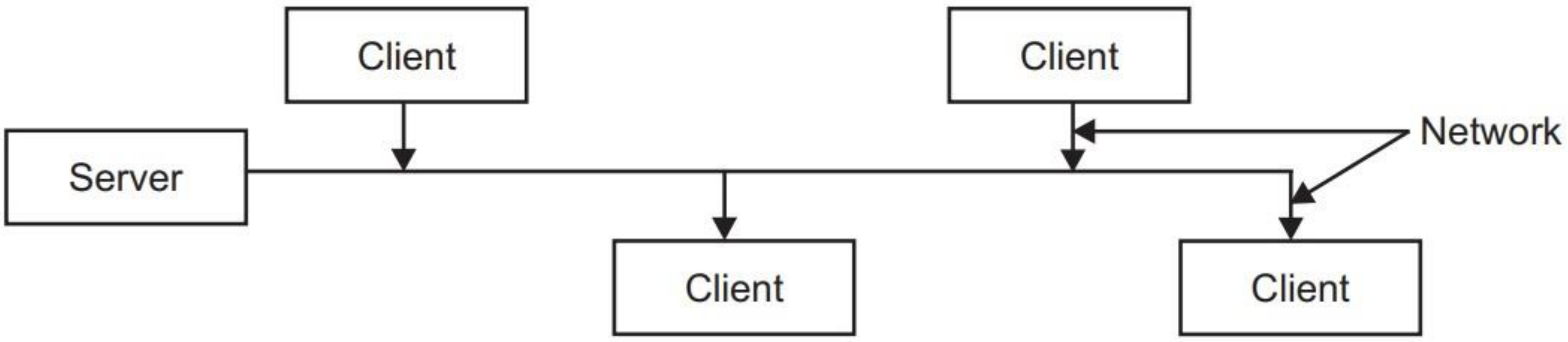

Figure 13: Client server database system

Components of Client-Server Architecture

There are three major components of client server architecture:

1. Server
2. Client
3. Network interface

1. Server :
   - Server is DBMS itself.
   - It consists of DBMS and supports all basic DBMS functions.
   - Server components of DBMS are installed at server.
   - It acts as monitor of all of its clients.

Functions of Server : The server performs various functions, which are as follows.

1. It supports all basic DBMS functions.
2. Monitor all his clients.
3. Distribute work-load over clients.
4. Solve problems which are not solved by clients.
5. Maintain security and privacy.
6. Avoiding unauthorized access of data.

2. Clients :

- Client machine is a personal computer or workstation which provide services to both server and users.
- It must obey his server.
- Client components of DBMS are installed at client site.
- Clients are taking instructions from server and help them by taking their load.
- When any user want to execute a query on client, the client first take data from server then execute the query on his own hardware and returns the result to the server.

3. Network Interface :

- Clients are connected to server by network interface.
- It is useful in connecting the server interface with user interface so that server can run his applications over his clients.

Comparison Between client/server and distributed database System

Client/Server Database System	Distributed Database System
1. In this, different platforms are often difficult to manage.	1. In this, different platforms can be managed easily.
2. Here, application is usually distributed across clients.	2. Here, application is distributed across sites.
3. In this database system, whole system comes to a halt if server crashes.	3. Here, failure of one site doesn't bring the entire system down as system may be able to reroute the one site's request to another site.
4. Maintenance cost is low.	4. Maintenance cost is much higher.
5. In this system, access to data can be easily controlled.	5. In DDS not only does the access to replicate the data has to be controlled at multiple locations but also the network has to be made secure.
6. In this, new sites can not be added easily.	6. In this, new sites can be added with little or no problem.
7. Speed of database access is good.	7. Speed of database access is much better.

Thank You!